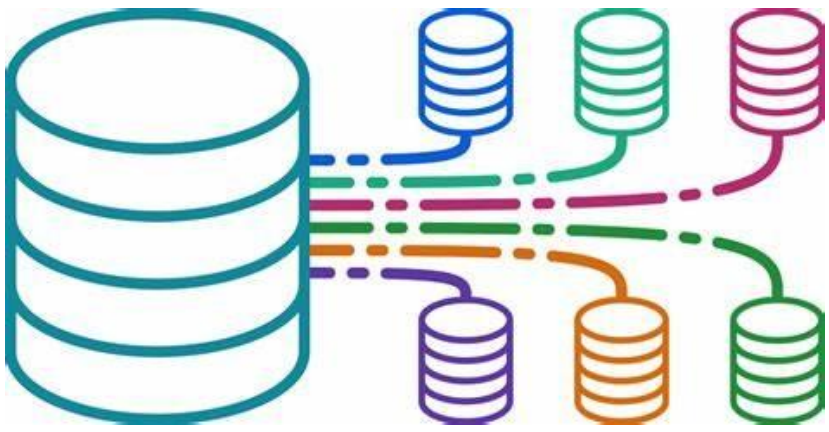
Remote Patient Monitoring Database System



Abdulmajeed Abdullah

Problem Description:

These days, we experience issues following a diagnosis and giving medication. When a patient visits a hospital, the attending physician examines and prescribes the necessary medication, and then asks the patient to return after a certain amount of time. When the patient returns, the physician examines the patient again, and in most cases, the examination results are good enough allowing the patient to leave. In this particular case, the patient Coming to see the doctor is expensive, both in terms of time and money, also the hospital needs to accommodate those who are most in need, From this point on, our idea started establishing a relationship between the patient and the doctor, as this relationship allows the doctor to take the patient's vital signs, assess his condition, and determine if the patient needs come to the him or not.

Background Information:

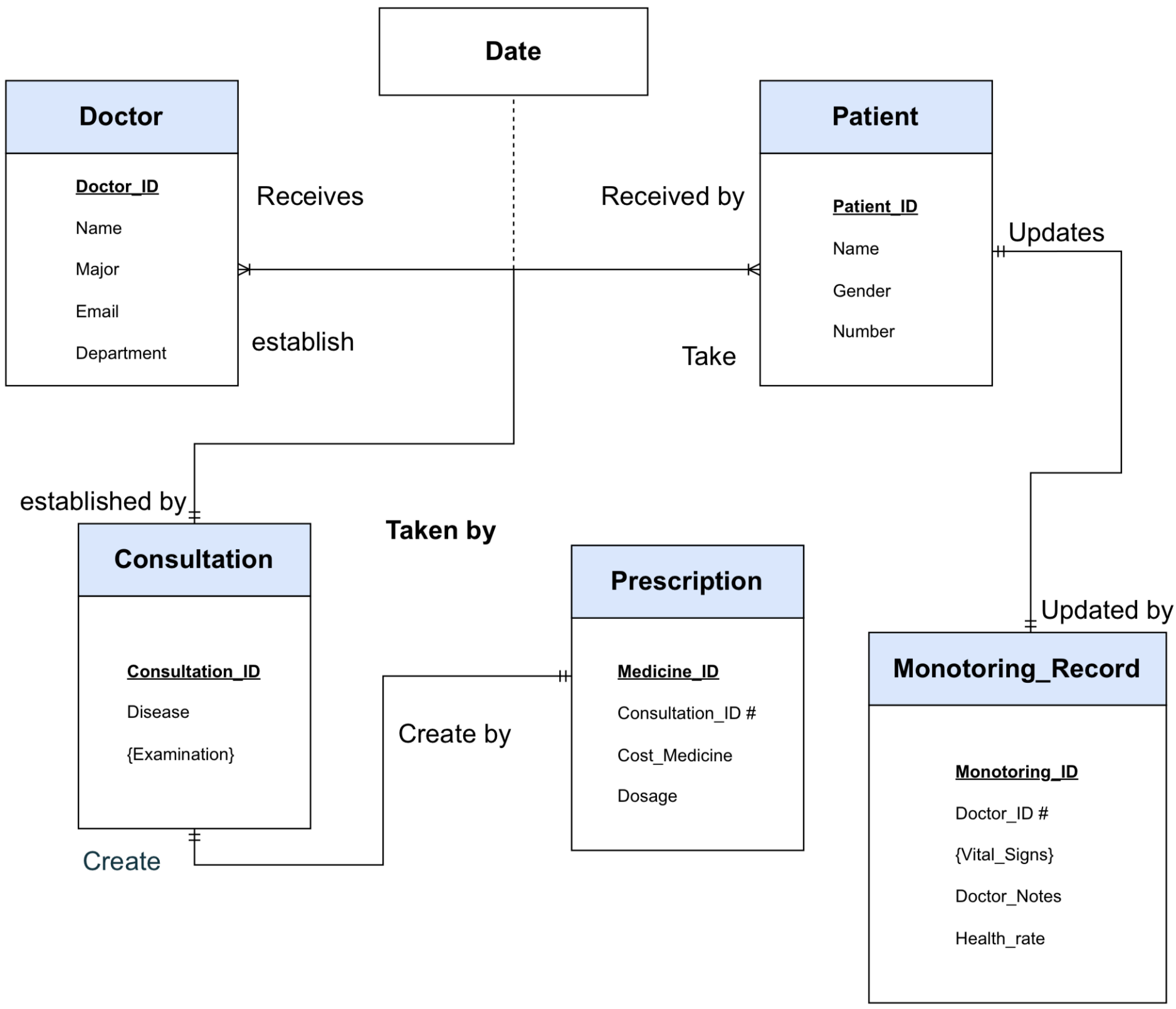
- The definition and benefits of remote patient monitoring (RPM), which is a technology that enables the collection and transmission of patient data from electronic devices to healthcare providers.
- The types of devices and conditions that can be monitored remotely, such as blood pressure, pulse oximetry, glucose, ECG, and more.
- The best practices and guidelines for implementing and managing an RPM program, such as device selection, patient enrollment, data analysis, care coordination, and reimbursement.
- The examples and case studies of successful RPM solutions and startups, such as CareCloud, ViveRPM, Health Recovery Solutions, and others.
- The current and future trends and challenges of RPM, such as regulatory issues, data security, patient engagement, and integration with other digital health solutions.
- The impact and outcomes of RPM on patient health, satisfaction, and cost savings, such as improved chronic disease management, reduced hospitalizations and readmissions, enhanced patient-provider communication, and lower healthcare expenditures .
- The potential barriers and risks of RPM, such as technical difficulties, device malfunction, data loss, inaccurate readings, patient privacy, and ethical dilemmas .
- The best practices and tips for patients who use RPM devices, such as how to set up and use the devices correctly, how to interpret and report the data, how to troubleshoot common problems, and how to communicate with their providers .
- The feedback and testimonials from patients and providers who have used RPM devices, such as how they felt about the experience, what they liked and disliked, what they learned, and what they would recommend to others .
- The comparison and evaluation of different RPM devices and platforms, such as their features, functions, costs, benefits, drawbacks, and user ratings .

Distribution of duties for the project:

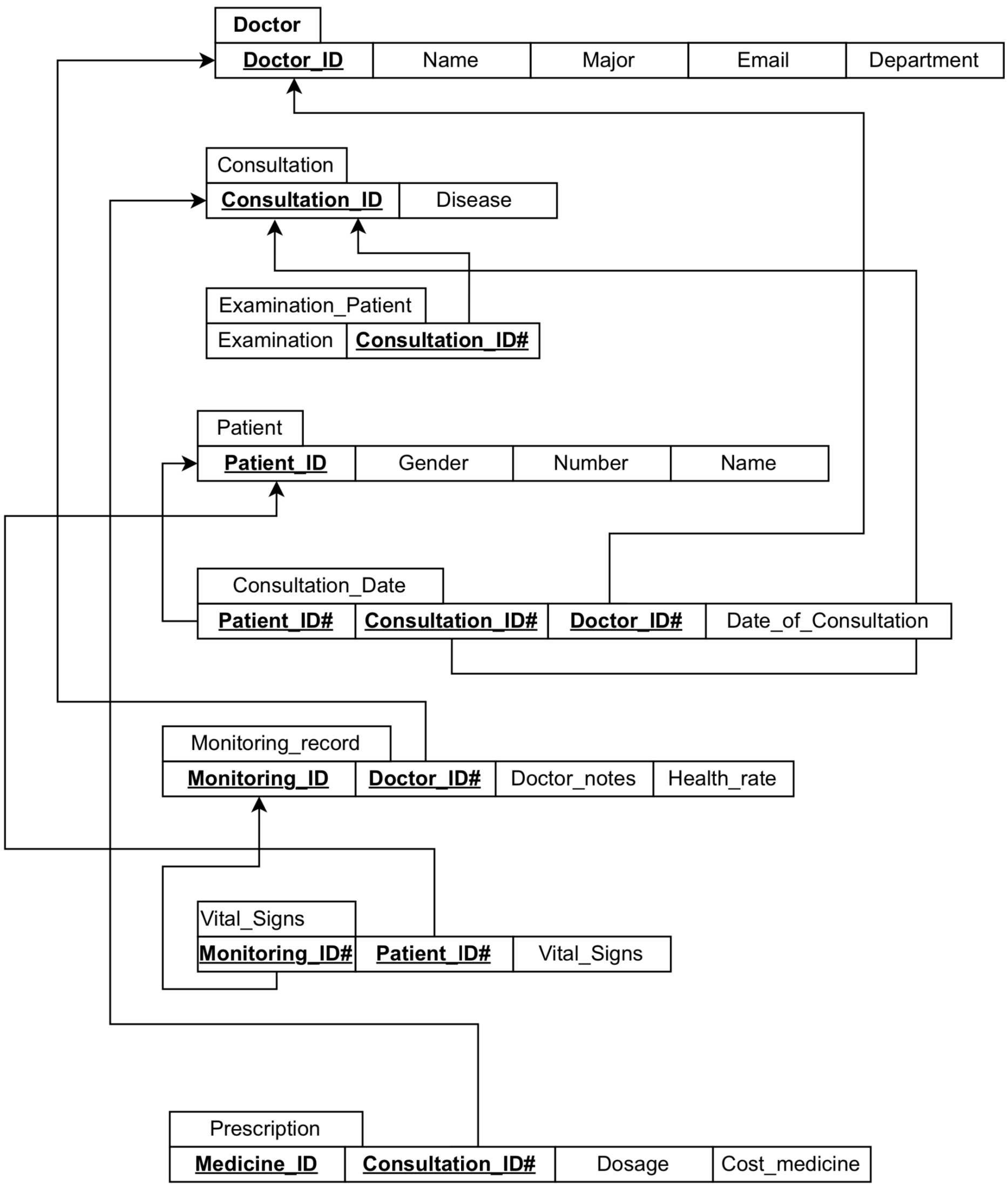
Initially, we all worked on the project but the most accurate division was split into two sections:

- The first is the theoretical section, where Abdulmajeed and Abdulmohsen are tasked with creating the project file, adding all the necessary data, designing the entities, mapping, and normalizing it.
- Second: The hands-on portion: Omar and Meshari are tasked with creating tables in MySQL, updating and adding data to them, making the queries with using Select, and creating stored procedures.

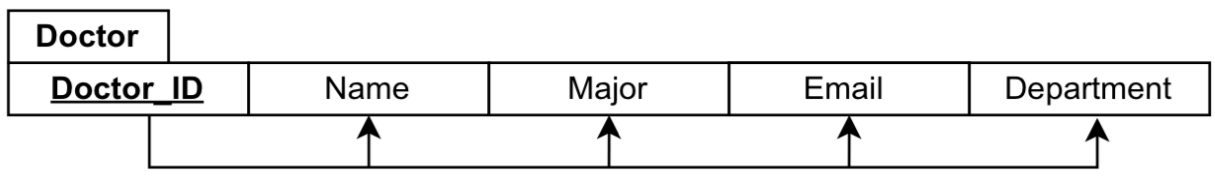
ER-Diagram:


Date
Doctor
Doctor_ID
Name
Major
Email
Department
Receives
establish
Received by
Take
Patient
Patient_ID
Name
Gender
Number
Updates
established by
Taken by
Consultation
Consultation_ID
Disease
{Examination}
Create
Create by
Prescription
Medicine_ID
Consultation_ID #
Cost_Medicine
Dosage
Updated by
Monotoring_Record
Monotoring_ID
Doctor_ID #
{Vital_Signs}
Doctor_Notes
Health_rate

Schema Diagram:



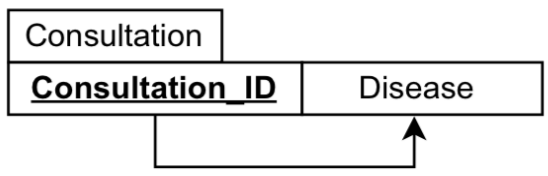
Normalization:



Functional Dependence:

Doctor_ID ⟶ Name, Major, Email, Department

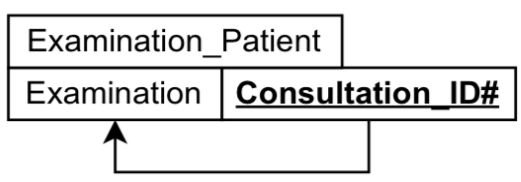
The relation in 3NF so it is also in 2NF and 1NF



Functional Dependence:

Consultation_ID ⟶ diseases

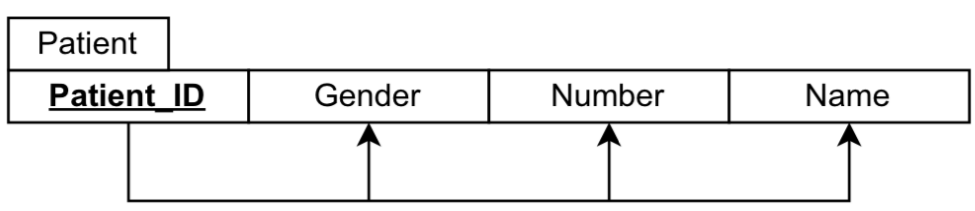
The relation in 3NF so it is also in 2NF and 1NF



Functional Dependence:

Consultation_ID ⟶ Examination

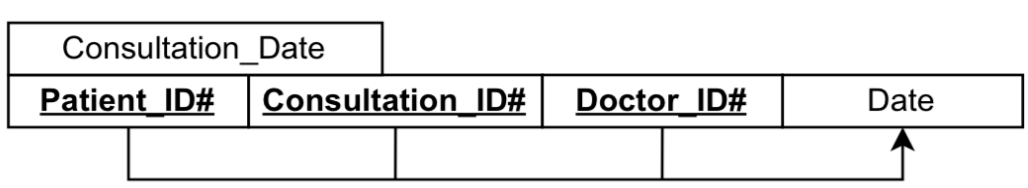
The relation in 3NF so it is also in 2NF and 1NF



Functional Dependence:

Patient_ID ⟶ Name, Gender, Number

The relation in 3NF so it is also in 2NF and 1NF



Functional Dependence:

Patient_ID, Consultation_ID, Doctor_ID ⟶ Date

The relation in 3NF so it is also in 2NF and 1NF

Monitoring_record			
Monitoring_ID	**Doctor_ID#**	Doctor_notes	Health_rate

Functional Dependence:

Monitoring_ID, Doctor_ID ⟶ Doctor_notes

Doctor_Notes ⟶ Health_rate

The relation in 2NF , we will convert it to 3NF:

Monitoring_record	**Monitoring_ID**	**Doctor_ID#**	Doctor_notes

Determination	DOCTOR_NOTES_ID #	Health_rate

Now the relation in 3NF

Vital_Signs		
Monitoring_ID#	**Patient_ID#**	Vital_Signs

Functional Dependence:

Monitoring_ID , Patient_ID ⟶ Vital_Signs

The relation in 3NF so it is also in 2NF and 1NF

Prescription			
Medicine_ID	**Consultation_ID#**	Dosage	Cost_medicine

Functional Dependence:

Medicine _ID, Consultation_ID ⟶ Dosage

Medicine_ID ⟶ Cost_medicine

The relation in 1NF , we will convert it to 2NF:

Prescription	**Medicine_ID**	**Consultation_ID#**	Dosage

Medicine	**Medicine_ID**	Cost_medicine

Now the relation in 2NF , also it is in 3NF

TABLES:

Statement 108

```
DESC Doctor
```

TABLE DOCTOR

Column	Null?	Type
DOCTOR_ID	NOT NULL	NUMBER
NAME	NOT NULL	VARCHAR2(100)
MAJOR	NOT NULL	VARCHAR2(100)
EMAIL	NOT NULL	VARCHAR2(200)
DEPARTMENT	–	VARCHAR2(30)

Download CSV

5 rows selected.

Statement 109

```
DESC Consultation
```

TABLE CONSULTATION

Column	Null?	Type
CONSULTATION_ID	NOT NULL	NUMBER
DISEASE	NOT NULL	VARCHAR2(150)

Download CSV

2 rows selected.

Statement 110

```
DESC Examination_Patient
```

TABLE EXAMINATION_PATIENT

Column	Null?	Type
EXAMINATION	NOT NULL	VARCHAR2(300)
CONSULTATION_ID	NOT NULL	NUMBER

Download CSV

2 rows selected.

Statement 111

```
DESC Patient
```

TABLE PATIENT

Column	Null?	Type
PATIENT_ID	NOT NULL	NUMBER
GENDER	NOT NULL	VARCHAR2(6)
PHONE_NUMBER	NOT NULL	CHAR(10)
NAME	NOT NULL	VARCHAR2(100)

Download CSV

4 rows selected.

Statement 112

```
DESC Consultation_Date
```

TABLE CONSULTATION_DATE

Column	Null?	Type
Patient_ID#	NOT NULL	NUMBER
Consultation_ID#	NOT NULL	NUMBER
Doctor_ID#	NOT NULL	NUMBER
DATE_OF_CONSULTATION	NOT NULL	DATE

Download CSV

4 rows selected.

Statement 113

```
DESC Monitoring_Record
```

TABLE MONITORING_RECORD

Column	Null?	Type
MONITORING_ID	NOT NULL	NUMBER
Doctor_ID#	–	NUMBER
Doctor_ Notes#	–	VARCHAR2(600)

Download CSV

3 rows selected.

Statement 114

```
DESC Vital_Signs
```

TABLE VITAL_SIGNS

Column	Null?	Type
MONITORING_ID	NOT NULL	NUMBER
VITAL_SIGNS	NOT NULL	VARCHAR2(300)
PATIENT_ID#	NOT NULL	NUMBER

Download CSV

3 rows selected.

Statement 115

```
DESC Prescription
```

TABLE PRESCRIPTION

Column	Null?	Type
MEDICINE_ID	NOT NULL	NUMBER
Consultation_ID#	NOT NULL	NUMBER
DOSAGE	NOT NULL	VARCHAR2(50)
MEDICINE_NAME	NOT NULL	VARCHAR2(40)
Patient#	–	NUMBER

Download CSV

5 rows selected.

Statement 116

```
DESC Medicine
```

TABLE MEDICINE

Column	Null?	Type
MEDICINE_ID	NOT NULL	NUMBER
COST_MEDICINE	NOT NULL	NUMBER(5,2)

Download CSV

2 rows selected.

Statement 117

```
DESC Determination
```

TABLE DETERMINATION

Column	Null?	Type
DOCTOR_NOTES_ID	NOT NULL	VARCHAR2(600)
HELTH_RATE	NOT NULL	VARCHAR2(400)

Download CSV

2 rows selected.

Inserted data:

```
Statement 16
INSERT INTO Doctor VALUES (1,'Ahmad Khaled','Cardiologist','ahmadkh@gmail.com','Cardiology Department')
1 row(s) inserted.

Statement 17
INSERT INTO Doctor VALUES (2,'Saleh Khaled','Ophthalmologist','saleh3kh@gmail.com','Ophthalmologist Department')
1 row(s) inserted.

Statement 18
INSERT INTO Doctor VALUES (3,'Talal Hamada','Orthopedist','tlal4h@gmail.com','Orthopedist Department')
1 row(s) inserted.

Statement 19
INSERT INTO Doctor VALUES (5,'Abeer Alrogy','Internist','abeer56@gmail.com','Internist Department')
1 row(s) inserted.

Statement 20
INSERT INTO Doctor (DOCTOR_ID,NAME,MAJOR,EMAIL) VALUES (4,'Wleed Baeshin','General Practitioner','wleed@gmail.com')
1 row(s) inserted.

Statement 23
INSERT INTO PATIENT  VALUES (1, 'Female', '0512345678', 'Asala Albogamy')
1 row(s) inserted.

Statement 24
INSERT INTO PATIENT  VALUES (2, 'Male', '0523456789', 'Mohammad Alwaqdany')
1 row(s) inserted.

Statement 25
INSERT INTO PATIENT  VALUES (3, 'Male', '0534567890', 'Aseel Alshihry')
1 row(s) inserted.

Statement 26
INSERT INTO PATIENT  VALUES (4, 'Female', '0545678901', 'Wedad Alsheek')
1 row(s) inserted.

Statement 27
INSERT INTO PATIENT  VALUES (5, 'Male', '0556789012', 'Ahmed Alyamy')
1 row(s) inserted.

Statement 29
INSERT INTO Consultation  VALUES (1001, 'Influenza')
1 row(s) inserted.

Statement 30
INSERT INTO Consultation  VALUES (1002, 'Cataract')
1 row(s) inserted.

Statement 31
INSERT INTO Consultation  VALUES (1003, 'Coronary Artery Disease')
1 row(s) inserted.

Statement 32
INSERT INTO Consultation  VALUES (1004, 'Hypertension')
1 row(s) inserted.

Statement 33
INSERT INTO Consultation  VALUES (1005, 'Type 2 Diabetes')
1 row(s) inserted.

Statement 38
INSERT INTO Examination_Patient  VALUES ( 'Administered annual flu vaccine and prescribed antiviral medication for Influenza',10)
1 row(s) inserted.

Statement 39
INSERT INTO Examination_Patient  VALUES ( 'Performed phacoemulsification with intraocular lens implant for Cataract removal',11)
1 row(s) inserted.

Statement 40
INSERT INTO Examination_Patient  VALUES ( 'Conducted cardiac catheterization to assess severity of Coronary Artery Disease',12)
1 row(s) inserted.

Statement 41
INSERT INTO Examination_Patient  VALUES ( 'Measured blood pressure and adjusted antihypertensive medication for Hypertension management',13)
1 row(s) inserted.

Statement 42
INSERT INTO Examination_Patient  VALUES ( 'Reviewed blood glucose records and optimized insulin therapy for Type 2 Diabetes',14)
1 row(s) inserted.
```

Statement 50

```
INSERT INTO Consultation_Date VALUES (1,1001,1,to_date('2023-11-10', 'YYYY-MM-DD'))

1 row(s) inserted.
```

Statement 51

```
INSERT INTO Consultation_Date VALUES (2,1002,2, to_date('2023-11-12', 'YYYY-MM-DD'))

1 row(s) inserted.
```

Statement 52

```
INSERT INTO Consultation_Date VALUES (3,1003,3,to_date('2023-11-16', 'YYYY-MM-DD'))

1 row(s) inserted.
```

Statement 53

```
INSERT INTO Consultation_Date VALUES (4,1004,5,to_date('2023-11-8', 'YYYY-MM-DD'))

1 row(s) inserted.
```

Statement 54

```
INSERT INTO Consultation_Date VALUES (4,1005,5,to_date('2023-11-15', 'YYYY-MM-DD'))

1 row(s) inserted.
```

Statement 58

```
INSERT INTO Determination VALUES ('Recommended rest, fluids, and over-the-counter fever reducers.
Consider antiviral medication if within the first 48 hours of symptom onset.', '70%')

1 row(s) inserted.
```

Statement 59

```
INSERT INTO Determination VALUES ('make sure you must not drive at night', '40%')

1 row(s) inserted.
```

Statement 60

```
INSERT INTO Determination VALUES ('Advised on lifestyle changes, including diet, exercise, and smoking cessation.', '55%')

1 row(s) inserted.
```

Statement 61

```
INSERT INTO Determination VALUES ('importance of lifestyle modifications such as reducing salt intake and regular physical activity.
Scheduled follow-up to monitor blood pressure and adjust medications as necessary','65%')

1 row(s) inserted.
```

Statement 62

```
INSERT INTO Determination VALUES ('importance of dietary management, regular blood glucose monitoring. and physical activitv.','58%')

1 row(s) inserted.
```

Statement 63

```
INSERT INTO Monitoring_Record VALUES (1,1,
   'Advised on lifestyle changes, including diet, exercise, and smoking cessation.')

1 row(s) inserted.
```

Statement 64

```
INSERT INTO Monitoring_Record VALUES (2,2,'make sure you must not drive at night')

1 row(s) inserted.
```

Statement 65

```
INSERT INTO Monitoring_Record (Monitoring_ID,"Doctor_ID#") VALUES (3,3)

1 row(s) inserted.
```

Statement 66

```
INSERT INTO Monitoring_Record VALUES (4,4, 'Recommended rest, fluids, and over-the-counter fever reducers.
Consider antiviral medication if within the first 48 hours of symptom onset.')

1 row(s) inserted.
```

Statement 67

```
INSERT INTO Monitoring_Record VALUES (5,5,'importance of dietary management, regular blood glucose monitoring. and physical activitv.')

1 row(s) inserted.
```

Statement 68

```
INSERT INTO Vital_Signs (Monitoring_ID, Vital_Signs)
VALUES (2, 'Pulse Rate: 70 bpm, Respiration Rate: 16 breaths/min, Blood Pressure: 118/78 mmHg, Blood Oxygen: 99%, Temperature: 36.6°C')

1 row(s) inserted.
```

Statement 69

```
INSERT INTO Vital_Signs (Monitoring_ID, Vital_Signs)
VALUES (3, 'Pulse Rate: 75 bpm, Respiration Rate: 18 breaths/min, Blood Pressure: 130/85 mmHg, Blood Oxygen: 95%, Weight: 85 kg')

1 row(s) inserted.
```

Statement 70

```
INSERT INTO Vital_Signs (Monitoring_ID, Vital_Signs)
VALUES (4, 'Pulse Rate: 80 bpm, Respiration Rate: 19 breaths/min, Blood Pressure: 150/90 mmHg, Blood Oxygen: 98%, Weight: 90 kg')
```

1 row(s) inserted.

Statement 71

```
INSERT INTO Vital_Signs (Monitoring_ID, Vital_Signs)
VALUES (5, 'Pulse Rate: 85 bpm, Respiration Rate: 20 breaths/min, Blood Pressure: 135/88 mmHg, Blood Oxygen: 96%, Blood Glucose Level: 180 mg/dl')
```

1 row(s) inserted.

Statement 72

```
INSERT INTO Vital_Signs VALUES (1,'Pulse Rate: 100 bpm, Respiration Rate: 20 breaths/min, Blood Pressure: 120/80 mmHg, Blood Oxygen: 97%, Temperature: 38.5°C')
```

1 row(s) inserted.

Statement 73

```
ALTER TABLE Prescription MODIFY Medicine_Name VARCHAR2 (40) NOT NULL
```

Table altered.

Statement 74

```
INSERT INTO Prescription  VALUES (1, 1001, 'Take one tablet every 6 hours', 'Oseltamivir')
```

1 row(s) inserted.

Statement 75

```
INSERT INTO Prescription VALUES (2, 1002, 'Use one drop in the affected eye twice a day', 'Prednisolone Acetate')
```

Statement processed.

Statement 76

```
INSERT INTO Prescription  VALUES (3, 1003, 'Take one tablet daily', 'Atorvastatin')
```

1 row(s) inserted.

Statement 77

```
INSERT INTO Prescription  VALUES (4, 1004, 'One tablet daily in the morning', 'Lisinopril')
```

1 row(s) inserted.

Statement 78

```
INSERT INTO Prescription VALUES (5, 1005, 'Inject 10 units subcutaneously before meals', 'Insulin Glargine')
```

1 row(s) inserted.

Statement 83

```
INSERT INTO Monitoring_Record VALUES (6,5,'importance of lifestyle modifications such as reducing salt intake and regular physical activity.
Scheduled follow-up to monitor blood pressure and adjust medications as necessary')
```

1 row(s) inserted.

Statement 90

```
INSERT INTO Vital_Signs VALUES (1,'Pulse Rate: 100 bpm, Respiration Rate: 20 breaths/min, Blood Pressure: 120/80 mmHg, Blood Oxygen: 97%, Temperature: 38.5°C',1)
```

1 row(s) inserted.

Statement 91

```
INSERT INTO Vital_Signs
VALUES (2, 'Pulse Rate: 70 bpm, Respiration Rate: 16 breaths/min, Blood Pressure: 118/78 mmHg, Blood Oxygen: 99%, Temperature: 36.6°C',2)
```

1 row(s) inserted.

Statement 92

```
INSERT INTO Vital_Signs
VALUES (3, 'Pulse Rate: 75 bpm, Respiration Rate: 18 breaths/min, Blood Pressure: 130/85 mmHg, Blood Oxygen: 95%, Weight: 85 kg',3)
```

1 row(s) inserted.

Statement 93

```
INSERT INTO Vital_Signs
VALUES (4, 'Pulse Rate: 80 bpm, Respiration Rate: 19 breaths/min, Blood Pressure: 150/90 mmHg, Blood Oxygen: 98%, Weight: 90 kg',4)
```

1 row(s) inserted.

Statement 94

```
INSERT INTO Vital_Signs
VALUES (5, 'Pulse Rate: 85 bpm, Respiration Rate: 20 breaths/min, Blood Pressure: 135/88 mmHg, Blood Oxygen: 96%, Blood Glucose Level: 180 mg/dl',4)
```

1 row(s) inserted.

Statement 102

```
INSERT INTO Medicine VALUES(1,327.78)
```

1 row(s) inserted.

Statement 103

```
INSERT INTO Medicine VALUES(2,150)
```

1 row(s) inserted.

Statement 104

```
INSERT INTO Medicine VALUES(3,41.25)
```

1 row(s) inserted.

Statement 105

```
INSERT INTO Medicine VALUES(4,30.41)
```

1 row(s) inserted.

Statement 106

```
INSERT INTO Medicine VALUES(5,483.75)
```

1 row(s) inserted.

Queries:

Statement 117

```
SELECT Name,Doctor_ID AS ID FROM Doctor WHERE Doctor_ID>2 ORDER BY Doctor_ID DESC
```

NAME	ID
Abeer Alrogy	5
Wleed Baeshin	4
Talal Hamada	3

Download CSV

3 rows selected.

Statement 118

```
SELECT Name,COUNT(Name),MIN(Patient_ID) FROM Patient GROUP BY Name HAVING MAX(Patient_ID)>2
```

NAME	COUNT(NAME)	MIN(PATIENT_ID)
Aseel Alshihry	1	3
Wedad Alsheek	1	4
Ahmed Alyamy	1	5

Download CSV

3 rows selected.

Statement 119

```
SELECT Name,Doctor_ID FROM Doctor WHERE Doctor_ID >(SELECT MIN(Patient_ID) FROM Patient)
```

NAME	DOCTOR_ID
Saleh Khaled	2
Talal Hamada	3
Wleed Baeshin	4
Abeer Alrogy	5

Download CSV

4 rows selected.

Statement 120

```
SELECT P.name FROM Patient P INNER JOIN Consultation_Date C ON P.Patient_ID = C."Patient_ID#" WHERE P.Patient_ID>2
```

NAME
Aseel Alshihry
Wedad Alsheek
Wedad Alsheek

Download CSV

3 rows selected.

Stored procedures:

Statement 104

```
CREATE OR REPLACE PROCEDURE doctorData(num Doctor.DOCTOR_ID%type)
  AS
  CURSOR data IS
  SELECT D.name ,D.doctor_id,D.major,D.email, D.department From doctor  D where  D.doctor_id=num;
  BEGIN
  FOR info IN data LOOP
  dbms_output.put_line('Data of Doctor: ID = '||info.Doctor_ID||',Name = '||info.Name|| ' ,Major = '||info.Major||' ,Email = '||info.Email||' ,Department = '||inf
  END LOOP;
  END doctorData;
```

```
Procedure created.
```

Statement 105

```
CREATE OR REPLACE PROCEDURE doctorData(num IN NUMBER)
AS
    CURSOR data IS
    SELECT name, doctor_id, major, email, department FROM doctor WHERE doctor_id=num;
BEGIN
    FOR info IN data LOOP
        dbms_output.put_line('Data of Doctor ID =' || info.doctor_id || '/Name =' ||
                             info.name || '/Major ='||info.major||'/Email ='||
                             info.email||'/Department ='||
                             info.department);
    END LOOP;
END doctorData;
```

```
Procedure created.
```

Statement 106

```
execute doctorData(1)
```

```
Statement processed.
Data of Doctor ID =1/Name =Ahmad Khaled/Major =Cardiologist/Email =ahmadkh@gmail.com/Department =Cardiology Department
```

Statement 121

```
CREATE OR REPLACE PROCEDURE DoctorNames(num Doctor.Doctor_ID%type)
 AS
CURSOR names IS
    SELECT Name,Doctor_ID FROM Doctor WHERE Doctor_ID >= num;
    BEGIN
    FOR Dname IN names LOOP
    dbms_output.put_line('Doctor name: '|| Dname.Name ||', Doctor_ID: '||Dname.Doctor_ID);
    END LOOP;
    END doctorNames;
```

```
Procedure created.
```

Statement 122

```
EXECUTE DoctorNames(4)
```

```
Statement processed.
Doctor name: Abeer Alrogy, Doctor_ID: 5
Doctor name: Wleed Baeshin, Doctor_ID: 4
```

Statement 123

```
CREATE OR REPLACE PROCEDURE setPatient (
    id Patient.Patient_id%type,
    new_Gender Patient.Gender%type,
    new_Phone_number Patient.phone_number%type,
    new_Name Patient.name%type
) AS

    old_Gender Patient.Gender%type;
    old_Phone_number Patient.phone_number%type;
    old_Name Patient.name%type;


BEGIN

    SELECT Gender,Phone_number,Name INTO old_Gender,old_Phone_number,old_Name FROM Patient WHERE Patient_id=id;
    dbms_output.put_line('Old Values: Gender = '||old_Gender||', Phone_number = '|| old_Phone_number|| ',Name = '||old_Name);


    UPDATE Patient
    SET Gender = new_Gender,
        Phone_number = new_Phone_number,
        Name = new_Name
    WHERE Patient_id = id;

   dbms_output.put_line('New Values: Gender = '||new_Gender||', Phone_number = '||new_Phone_number||', Name = '||new_Name);

END setPatient;
```

```
Procedure created.
```

Statement 124

```
EXECUTE setPatient(2,'male','0500000000','Mohammad Alwaqdany')
```

```
Statement processed.
Old Values: Gender = Male, Phone_number = 0523456789,Name = Mohammad Alwaqdany
New Values: Gender = male, Phone_number = 0500000000, Name = Mohammad Alwaqdany
```